NAJIYA OMAR

PhD, P.Eng

CONTACT

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74 Fleetview Dr, Halifax, Nava Scotia

ACADEMIC EDUCATION

Doctor of Philosophy: Electrical and Computer Engineering
Dalhousie University | Halifax, NS | 2023

Master of Applied Science: Electrical and Computer Engineering
Dalhousie University | Halifax, NS | 2017

Bachelor's in electrical and computer engineering with Distinction
Sirte University | Sirte, Libya | 2004

PROFESSIONAL CERTIFICATION

Professional Engineers (P.Eng)
Association of Engineers Nova Scotia |
Halifax, NS |2024

SKILLS

➢ TECHNICAL

- Python for Statistical Analysis, Data Science and Machine Learning
- Power BI, Tableau
- Power system simulation software (PSS/E)

PROFESSIONAL SUMMARY

I am an experienced electrical engineer with a passion for leveraging big data for energy management system transformation. My area of expertise is in predictive analysis, optimization theory, and advanced technologies to create smart energy management solutions. With a deep understanding of energy consumption patterns, renewable energy sources, and cutting-edge optimization techniques, I am interested in developing scalable technologies that enhance efficiency, reduce costs, and minimize environmental impact. My work involves integrating solar and other renewable resources to deploy machine learning algorithms for predictive modeling and system optimization.

PROFESSIONAL EXPERIENCE

RESEARCH FELLOW

Green Power Labs Inc , Dartmouth, NS | January 2022 – December 2023

- Optimized the data downloads by writing a code utilizing "GOES-2-Go" and s3fs python libraries to access the NOAA's GOES-16 geostationary satellite dataset that is stored as netCDF files in Amazon Web Services (AWS)/ S3 Bucket.
- Automated downloaded GOES-16 satellites images from different channels of the Advanced Baseline Imagery (ABI) instrument with different temporal and spatial resolutions, and map projections.
- Utilized the GOES-16 cloud products including Cloud Optical Depth, Clear Sky Mask , Cloud-top products such as Cloud-Top Height, Cloud-Top Pressure , Cloud-Top Temperature, Cloud Top Phase, and Cloud Particle Size for proposing Could Index (CI) dataset.
- Applied a map projection that positioned all the extracted variables from the ABI cloud products to the same projected grids.
- Selected appropriate thresholds and computing measurements for CI and prepared the obtained CI measurements to be used by the CI predictive algorithm.
- Employed the PVlib python package for calculating cloud irradiance components.

RESEARCH ASSISTANT

| Nova Scotia Community College, Dartmouth | January 2018 - May 2018

- -Developed and implemented experimental designs for data collection in Proteus DS simulation, utilizing skills in Matlab, resulting in improved accuracy and efficiency in data retrieval.
- Designed and implemented visualization solutions for inconsistencies in Remotely Operated Vehicles (ROV) simulation data, providing expertise in cleaning, organizing, and combining multivariate data for maximum utility.
- Conducted complex analyses of ROV data to determine feasible speed ranges for safe marine operations and determine capacity requirements of ROV thrusters, utilizing advanced graphing techniques and data visualization tools.
- Collaborated closely with team members, taking feedback and implementing changes to improve tool functionality and usability.
- Designed and maintained research databases and information systems, ensuring data integrity and accessibility for use in future research projects.
- Contributed to the publication of research findings in academic journals and conferences, presenting research results to a range of stakeholders, including industry professionals and academic peers.

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SKILLS

➢ PROFESSIONAL

- Exceptional organizational, analytical, and time-management skills
- Excellent interpersonal skills
- Works well independently, collaboratively, and in leadership roles

PROFESSIONAL TRAINING AND SEMINARS

Python for Time Series Analysis and Forecasting | April 2021

Programing for AI | NS
April 2025

EXPERIENCE CONTINUED...

TEACHING ASSISTANT

Dalhousie University | Halifax | September 2015 – December 2021

- Provided instruction and guidance to undergraduate students in Electric Circuits and Computer Programming courses, using hands-on examples and clear communication skills to provide one-to-one support.
- Demonstrated laboratory instruments, including prototype boards, oscilloscopes, and digital multimeters, to first-time users, ensuring an understanding of theoretical concepts and safe laboratory practices.
- Marked and corrected assignments and lab reports, providing feedback on errors in terms of principle electrical theories and programming concepts.
- Performed troubleshooting skills to help students understand C and C++ codes, identifying and resolving problems in the execution of code.
- Supervised the work of 30-40 undergraduate students, providing constructive feedback to help improve their performance.
- Invigilated and marked final tests, providing detailed reports to department staff for grading and record-keeping purposes.

PROGRAMMER AND ELECTRICAL ENGINEER

AL-Mofida Construction Company | Sirte,Libya| May 2009 – May 2013

- Collaborated with a multidisciplinary team to successfully deliver government-funded energy projects with a budget of approximately $500,000, playing a key role in the projects' success.
- Leveraged sophisticated Artificial Neural Network (ANN) applications to accurately predict and estimate power consumption for government energy projects in Sirte, using advanced statistical metrics to evaluate algorithm performance and ensure accuracy rates that met or exceeded project requirements.
- Applied Machine Learning (ML) algorithms to classify datasets for pattern recognition tasks, using best practices in data management to streamline and optimize the data analysis process.
- Attended regular project meetings to provide up-to-date insights and outcomes, and skillfully managed the financial plan for each project, optimizing resource allocation and budgeting to meet project goals.

REFERENCES AVAILABLE ON REQUEST